

Chronological Networks as a Library Primitive: ChronoLog over an Interval-Native Date/Time Stack

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Independent

Abstract

Levy, Geeraerts, Pluquet, Piasetzky and Fantalkin (2020) formalised archaeological chronology as a *chronological network* — a set of time-periods related by synchronisms, asynchronisms, and duration bounds — and showed that consistency checking and the computation of tightest possible date ranges reduce to a Simple Temporal Problem (STP), solvable in polynomial time. Their tool, ChronoLog, invents an interval/relation vocabulary from scratch. We observe that this vocabulary is exactly the one a general, interval-native date/time library already provides: half-open intervals, Allen’s thirteen relations, partial and reduced-precision dates, calendar-aware durations, and uncertainty qualifiers. We implement the ChronoLog model as a small layer (**Tempo.Network**) over the Tempo Elixir library, mapping each ChronoLog relation to its Allen counterpart plus metric (delay) extensions, normalising a network to an STP, and solving consistency and tightening by Floyd–Warshall. The layer reproduces the paper’s published bounds exactly and inherits, for free, ISO 8601 / EDTF interchange, multiple calendars, BCE arithmetic, and §8 uncertainty qualifiers. The contribution is less a new algorithm than an argument: chronological-network modelling is a natural *library primitive* when the library already treats time as intervals.

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Related Version Source repository: <https://github.com/kipcole9/tempo>

1 Introduction

Archaeological and historical chronology is rarely a list of fixed dates. It is a web of relationships: a king reigns before another, a stratum is built during a reign, a ceramic style overlaps its successor, an inscription supplies a *terminus post quem*. Levy et al. [6] call such a web a *chronological network* and give it a formal model with three object types — time-periods, sequences, and chronological relations — and two operations: *consistency checking* (does any assignment of dates satisfy every constraint?) and *tightening* (what is the narrowest start, end, and duration each period can take?). Their key technical result is that the model is a Simple Temporal Problem [2]: every relation, duration, and absolute date reduces to atomic constraints of the form $b_1 - b_2 \leq k$ over boundary variables, so consistency is the absence of a negative cycle and tightening is all-pairs shortest paths. This is polynomial-time, in contrast to exhaustive-search tools such as Groundhog [3].

ChronoLog [7] implements this model as a bespoke application, and in doing so re-creates a small interval algebra: a period has a start, an end, and a duration, each possibly unknown, bounded, or ranged; relations express containment, overlap, synchronised boundaries, and metric delays. Our observation is that this is precisely the vocabulary that an *interval-native* general-purpose date/time library already provides. We demonstrate this with Tempo, an Elixir library in which every temporal value is a bounded half-open interval at a declared



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45 resolution, comparison is Allen’s thirteen relations [1], and partial ISO 8601 / EDTF values
 46 (2022, 2022-06, ~720) are first-class.

47 We contribute **Tempo.Network**, a layer that (i) names the ChronoLog relation vocabulary
 48 and maps it to Allen’s algebra plus metric extensions (§4); (ii) normalises a network to
 49 an STP against a single origin boundary (§5); and (iii) solves consistency and tightening
 50 by Floyd–Warshall, with shortest-path *traces* that explain each derived bound (§5). The
 51 layer reproduces the paper’s ChronoLand and 26th-dynasty results exactly (§6) and inherits
 52 ISO 8601/EDTF interchange, calendars, and qualifiers from the host library (§7).

53 2 The ChronoLog model

54 A *time-period* is a continuous interval — a reign, an era, the life of a stratum — characterised
 55 by a start date, an end date, and a duration, each of which may be unknown, known,
 56 lower-bounded, upper-bounded, or known within a range $[lo, hi]$. Dates and durations are
 57 modelled *independently*: a period is at most six numbers (earliest/latest start, earliest/latest
 58 end, minimum/maximum duration), reconciled by the relation $duration = end - start$.

59 A *sequence* is a list of consecutive time-periods with no gaps: $end(p_i) = start(p_{i+1})$. A
 60 gap is modelled by inserting an explicit gap period.

61 A *chronological relation* constrains two periods. The vocabulary (the paper’s Tables 1–5)
 62 covers contemporaneity ($A \sim B$: non-empty overlap), inclusion ($A \subseteq B$, $A \supseteq B$), overlap
 63 ($A \leq B$, $A \geq B$), start/end-period synchronisms (“A starts during B”, “A includes the start
 64 of B”, and the end-side duals), synchronised boundaries (synchronous start/end, equality,
 65 immediate precedence/following), asynchronisms ($A \ll B$, $A \gg B$), ordered boundaries
 66 (strict orderings of individual start/end dates), and *delay synchronisms* — metric constraints
 67 stating that one boundary is an exact, minimum, or maximum number of years before or
 68 after another.

69 2.0.0.1 The STP reduction.

70 Every relation, duration, sequence link, and absolute date normalises to a conjunction of
 71 atomic constraints of one shape, $b_1 - b_2 \leq k$, where b_1, b_2 are boundary variables and k is
 72 an integer. Equalities become two inequalities; a strict $<$ on an integer time-scale becomes
 73 $\leq k - 1$; absolute dates are constraints against a single origin boundary z_0 . The constraints
 74 form a directed weighted graph with one node per boundary plus z_0 . The network is *consistent*
 75 iff the graph has no negative cycle, and the tightest bound on $b_1 - b_2$ is the shortest-path
 76 weight $b_1 \rightarrow b_2$; tightening is therefore all-pairs shortest paths (Floyd–Warshall) [2, 4].

77 3 Tempo’s interval model

78 Tempo represents every temporal value as a bounded, half-open interval $[from, to)$ at a
 79 declared resolution. A partial ISO 8601 value is an interval of the appropriate width:
 80 2026 is $[2026-01-01, 2027-01-01)$, 2026-06 spans one month, 2026-06-15 one day. The
 81 half-open convention makes adjacent spans concatenate without gap or overlap, and makes
 82 comparison total under Allen’s thirteen relations, exposed as predicates (**before?**, **overlaps?**,
 83 **within?**, **adjacent?**). Durations are calendar-aware (P20Y, P2Y6M), and ISO 8601-2 / EDTF
 84 uncertainty qualifiers (? uncertain, ~ approximate, % both) are parsed and carried per
 85 component.

86 These are exactly the primitives ChronoLog needs. A time-period’s boundaries are
 87 interval endpoints; a sequence is the half-open concatenation $[a, b)$ then $[b, c)$; the relations

■ **Table 1** Selected ChronoLog relations, their Allen counterpart, and their atomic constraints. s, e abbreviate start, end.

ChronoLog relation	Allen	Atomic constraints
$A \ll B$ (before)	precedes	$e(A) - s(B) \leq -1$
$A \sqsupset B$ (immed. prec.)	meets	$e(A) = s(B)$
$A \subseteq B$ (included in)	during	$s(B) \leq s(A), e(A) \leq e(B)$
$A \supseteq B$ (includes)	contains	$s(A) \leq s(B), e(B) \leq e(A)$
$A \leq B$ (overlaps)	overlaps	$s(A) \leq s(B) \leq e(A) \leq e(B)$
$A \sim B$ (contemporary)	—	$s(B) \leq e(A), s(A) \leq e(B)$
A starts during B	—	$s(B) \leq s(A) \leq e(B)$
A ends during B	—	$s(B) \leq e(A) \leq e(B)$
$A \top B$ (synchronous start)	—	$s(A) = s(B)$
delay: $s(A)$ at least X before $s(B)$	—	$s(A) - s(B) \leq -X$

are Allen relations plus metric offsets; absolute and floating chronologies are anchored and unanchored values; uncertainty is the qualifier vocabulary. The one missing piece is the constraint solver, which is small and well-understood. **Tempo.Network** supplies it.

4 Mapping the relation vocabulary

Each ChronoLog relation is stored as data and translated, by `to_atomic/1`, to its atomic constraints; Table 1 gives the correspondence to Allen’s algebra and the atomic form. Where a single Allen relation exists the mapping is one-to-one (and invertible, so a solved network can be *queried* with Allen predicates and an Allen relation between two solved periods can be *named*). The period synchronisms (“starts during” and *kin*) are looser than any single Allen relation — they are disjunctions — and the metric delays carry an integer Allen cannot express.

5 Normalisation and solver

Tempo.Network.Normalize reduces a network to $\{nodes, edges, unit\}$. Each period contributes `start` and `end` nodes; a shared origin `:origin(z0)` anchors absolute dates. The network’s *finest unit* — the finest resolution among its dates — fixes the integer axis: a date becomes its count of that unit relative to z_0 , a duration its count of that unit. Conversion is exact for the uniform-unit (typically year-grained) chronologies that are the discipline’s norm; a duration expressed across units uses nominal calendar lengths. Each edge is tagged with the constraint that produced it, for tracing.

Tempo.Network.Solver runs Floyd–Warshall once, in $O(n^3)$ on the boundary count — interactive for the hundreds of periods these chronologies contain. `consistent?/1` reports the absence of a negative cycle; `tighten/1` reads the tightest start, end, and duration of each period off the shortest-path matrix and maps them back to Tempo values. `trace/3` reconstructs the shortest path underlying a chosen bound and renders it as prose — the paper’s notion of an explanatory trace, where every step names the input it rests on.

113 6 Worked examples

114 6.0.0.1 ChronoLand.

115 The paper’s toy kingdom: kings K1 then K2 reign in succession, both between 1200 and
 116 1300 CE; K1 reigns at most 10 years, K2 at least 35. Strata S1 (built by K1) and S2 (destroyed
 117 under K2) follow one another, each lasting 20–100 years.

```
118 network =
119     Network.new()
120     |> Network.add_period(:k1, start: {:not_before, ~o"1200Y"}, duration: {:at_most, ~o"P10Y"})
121     |> Network.add_period(:k2, end: {:not_after, ~o"1300Y"}, duration: {:at_least, ~o"P35Y"})
122     |> Network.add_period(:s1, duration: {~o"P20Y", ~o"P100Y"})
123     |> Network.add_period(:s2, duration: {~o"P20Y", ~o"P100Y"})
124     |> Network.add_sequence([:k1, :k2])
125     |> Network.add_sequence([:s1, :s2])
126     |> Network.add_relation(:starts_during, :s1, :k1)
127     |> Network.add_relation(:ends_during, :s2, :k2)
```

128 Tightening reproduces the paper’s Fig. 6b exactly: K2 tightens to $\text{start} \in [1200, 1265]$,
 129 $\text{end} \in [1240, 1300]$, $\text{duration} \in [35, 100]$; the strata, given no dates as input, acquire them,
 130 and each stratum’s duration tightens from $[20, 100]$ to $[20, 80]$ because the dynasty is too
 131 short to hold a longer one. The trace for the earliest end of K2 reproduces the paper’s Fig. 6c
 132 reasoning: K1 starts no earlier than 1200; S1 starts during K1; S1 lasts at least 20 years, so
 133 ends after 1220; S1 meets S2; S2 lasts at least 20 years, so ends after 1240; S2 ends during K2,
 134 so K2 ends after 1240. Capping K2 at 25 rather than 35 years makes the network inconsistent
 135 (Fig. 7), detected as a negative cycle.

136 6.0.0.2 The 26th dynasty.

137 Six reigns in succession (after Kitchen 2000). Given only Psammetichus I’s accession
 138 (664 BCE) and each reign’s length, tightening derives every reign’s absolute dates — exercising
 139 BCE (negative) years and a six-period sequence. Amasis II’s accession is recovered as 570 BCE,
 140 forced by the anchor and the five preceding reigns, with no date stated for it directly.

141 7 What the library adds, and its limits

142 Because boundaries are ordinary Tempo values, a network is interoperable by construction:
 143 each period round-trips through ISO 8601 / EDTF for exchange with other tooling, BCE and
 144 proleptic dates are handled uniformly, and non-Gregorian calendars are available through the
 145 underlying calendrical stack. Uncertainty qualifiers are carried as metadata and, by design,
 146 do *not* silently widen a bound: a vague “circa” never performs arithmetic unless the modeller
 147 restates it as a range. The solver shares interval-arithmetic primitives with the library’s
 148 planned set-operations milestone (union, intersection, coalescing of interval sets).

149 7.0.0.1 Limitations.

150 Following the source model, relations compose by *conjunction*; a genuine *or*-relation (“A ends
 151 before B *or* A starts after B”) is outside the STP and is not supported. There is no fuzzy
 152 or Bayesian layer; uncertainty is bounds, not probability. Cross-unit duration conversion

(e.g. a year-grained duration on a day axis) uses nominal calendar lengths and is therefore approximate, though the uniform-unit chronologies that dominate practice are exact.

8 Related work

Allen’s interval algebra [1] underlies the relation vocabulary; Holst [5] applied temporal logic to archaeology. The direct antecedent is ChronoLog [6, 7]; the STP machinery is that of Dechter, Meiri and Pearl [2] via Floyd [4]. Falk’s Groundhog [3] addresses the same problem by exhaustive search. Tempo’s contribution is orthogonal to all of these: it shows that, given an interval-native host library, the network model and its solver are a compact, reusable primitive rather than a standalone application.

9 Conclusion

Chronological-network modelling does not require a bespoke tool. When the host library already represents time as half-open intervals with Allen’s algebra, partial dates, and calendar-aware durations, the ChronoLog model is a thin layer: a relation-to-atomic translation, an STP normalisation, and a Floyd–Warshall solver. The result reproduces the reference results exactly while inheriting ISO 8601/EDTF interchange, calendars, and uncertainty qualifiers — and makes consistency checking and tightening available wherever the library is.

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